

Proofs

- **Exercise 1**

Let $a, b \in \mathbb{R}$ that satisfy $a + b = 10$. Prove that $ab \leq 25$ or $a^2 + b^2 < 10$.

- **Solution 1.1**

Suppose $a, b \in \mathbb{R}$ satisfy $a + b = 10$, then $a = 10 - b$, and we have

$$\begin{aligned} ab &= (10 - b)b \\ &= 10b - b^2 \\ &= -(b^2 - 10b) \\ &= -(b^2 - 10b + 25) + 25 \\ &= -(b - 5)^2 + 25 \\ &\leq 25. \end{aligned}$$

So $ab \leq 25$, then $ab \leq 25$ or $a^2 + b^2 < 10$.

- **Exercise 2**

Prove that $(\forall x) P(x) \vee (\forall x) Q(x) \implies (\forall x) (P(x) \vee Q(x))$ and refute the converse implication.

- **Solution 2.1**

Suppose $(\forall x) P(x) \vee (\forall x) Q(x)$ is true, then $(\forall x) P(x)$ is true or $(\forall x) Q(x)$ is true.

Suppose $(\forall x) P(x)$ is true, then $\forall x, P(x)$ is true, then $\forall x, P(x) \vee Q(x)$ is true and then $\forall x, (P(x) \vee Q(x))$ is true.

To check that the converse is false, consider $P(x) : x$ is even, $Q(x) : x$ is odd.

- **Exercise 3**

Proof: $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$

- **Solution 3.1**

We want to prove $\forall a, b, c \in \mathbb{R}, \min\{a, \max\{b, c\}\} = \max\{\min\{a, b\}, \min\{a, c\}\}$. Let

$$\begin{aligned} M_1 &:= \min\{a, \max\{b, c\}\}. \\ M_2 &:= \max\{\min\{a, b\}, \min\{a, c\}\}. \end{aligned}$$

we want to prove $M_1 \leq M_2$ and $M_2 \leq M_1$.

For the case $M_1 \leq M_2$, by the definition of min and max

$$\begin{aligned} M_1 &\leq a \wedge M_1 \leq \max\{b, c\} \\ \implies M_1 &\leq a \wedge (M_1 \leq b \vee M_1 \leq c) \\ \implies (M_1 &\leq a \wedge M_1 \leq b) \vee (M_1 \leq a \wedge M_1 \leq c) \\ \implies M_1 &\leq \min\{a, b\} \vee M_1 \leq \min\{a, c\} \\ \implies M_1 &\leq \max\{\min\{a, b\}, \min\{a, c\}\} \\ \implies M_1 &\leq M_2. \end{aligned}$$

For the case $M_2 \leq M_1$, by the definition of min and max, we have

$$\min\{a, b\} \leq a, \min\{a, b\} \leq b \leq \max\{b, c\}.$$

then $\min\{a, b\} \leq \min\{a, \max\{b, c\}\}$.

The other inequality is almost the same. Then $M_2 = \max\{\min\{a, b\}, \min\{a, c\}\}$.

- **Solution 3.2**

We want to prove $\forall a, b, c \in \mathbb{R}, \min\{a, \max\{b, c\}\} = \max\{\min\{a, b\}, \min\{a, c\}\}$.

Suppose $a, b, c \in \{0, 1\}$. We split into two cases depending on the value of a .

Case 1 $a = 0$

$$\text{LHS} = \min \{0, \max \{b, c\}\} = 0.$$

$$\text{RHS} = \max \{\min \{0, b\}, \min \{0, c\}\} = \max \{0, 0\} = 0.$$

Thus $\text{LHS} = \text{RHS}$.

Case 2 $a = 1$

$$\text{LHS} = \min \{1, \max \{b, c\}\} = \max \{b, c\}.$$

$$\text{RHS} = \max \{\min \{1, b\}, \min \{1, c\}\} = \min \{b, c\}.$$

Thus $\text{LHS} = \text{RHS}$.

Since equality holds for $a = 0$ and $a = 1$, the identity is true for all $a, b, c \in \{0, 1\}$.

• **Exercise 4**

Let $a_1, a_2, \dots, a_7 \in \mathbb{Z}$, prove that among these seven integers, one can always choose four numbers whose sum is even.

◦ **Solution 3.1**

Suppose $a_1, a_2, \dots, a_7 \in \mathbb{Z}$. Let K be the number of odd integers among $a_1, a_2, \dots, a_7$.

If $4 \leq K \leq 7$, then take 4 of these odd integers, their sum is even:

$$(2a + 1) + (2b + 1) + (2c + 1) + (2d + 1) = 2(a + b + c + d + 2).$$

If $0 \leq K < 4$, then take 4 of these even integers, their sum is even:

$$2a + 2b + 2c + 2d = 2(a + b + c + d).$$

In both cases, we obtain that we can always choose four numbers whose sum is even.

• **Exercise 5**

Prove that $\forall n \in \mathbb{Z}$, if n is odd, then $3n$ is odd.

◦ **Solution 5.1**

Suppose $n \in \mathbb{Z}$, and n is odd. Then, there exists an integer $k \in \mathbb{Z}$ such that $n = 2k + 1$. Then

$$\begin{aligned} 3n &= 3(2k + 1) \\ &= 6k + 3 \\ &= 2(3k + 1) + 1. \end{aligned}$$

Since $3k + 1$ is an integer, $3n$ is odd.

• **Exercise 6**

For every $a, b, c \in \mathbb{Z}$, prove that if $a \mid b$ and $b \mid c$, then $a \mid c$.

◦ **Solution 6.1**

Suppose $a, b, c \in \mathbb{Z}$ and $a \mid b \wedge b \mid c$. Then there exist $k, q \in \mathbb{Z}$ such that $b = ka$ and $c = qb$. Then $c = qka$, and since $qk \in \mathbb{Z}$, by definition, $a \mid c$.

• **Exercise 7**

Let $n \in \mathbb{Z}$. Prove that if n^2 is even, then n is even.

◦ **Solution 7.1**

We will prove the contrapositive: $\forall n \in \mathbb{Z}$, if n is odd, then n^2 is odd.

Suppose $n \in \mathbb{Z}$ is odd. Then $\exists k \in \mathbb{Z}$ such that $n = 2k + 1$. Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

is odd because $2k^2 + 2k$ is an integer.

- **Exercise 8**

Prove that there is no smallest positive rational number.

- **Solution 4.7.1**

By contradiction, suppose there exists a smallest positive rational number r . Then $\frac{r}{2}$ is a smaller positive rational number. This is a contradiction because r was assumed to be the smallest. Thus, there doesn't exist a smallest positive rational number.

- **Exercise 9**

$\forall \epsilon > 0$, prove that $\exists N \in \mathbb{N}$ such that $\forall n \geq N, \frac{1}{n} < \epsilon$.

- **Solution 9.1**

Suppose $\epsilon > 0$. We will first prove that $\exists N \in \mathbb{N}$ such that $\frac{1}{N} < \epsilon$.

The Archimedean Property states that for any real number $x \in \mathbb{R}$, there exists a natural number $N \in \mathbb{N}$ such that $N > x$. By setting $x = \frac{1}{\epsilon}$ in the Archimedean Property, we guarantee that $\exists N \in \mathbb{N}$ satisfying $N > \frac{1}{\epsilon}$, and then $\epsilon > \frac{1}{N}$. Then if $n \geq N$, $\frac{1}{N} \geq \frac{1}{n}$ and $\frac{1}{n} < \epsilon$.